

A Universe in a Bottle — Claims and Falsifiers

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One-page summary

A deterministic **quantum cellular automaton (QCA)** on a body-centred-cubic (BCC) lattice is proposed as the physical vacuum. The free dynamics are *forced* (not chosen) by the Bisio-D’Ariano-Perinotti-Tosini uniqueness theorem: in 3D the minimal non-trivial one-particle QCA is the Weyl walk on the BCC lattice. From one reinterpretation — **the speed of light is the angular rotation rate of a real $(\mathbf{E}, \mathbf{B})$ pair per unit wavenumber**, $c_{\text{lat}} = 1/\sqrt{3}$ — the model recovers Maxwell’s equations (as a linearised rotation), the Einstein mass shell (as a spherical-Pythagorean identity), and, with gauge structure added, the Standard-Model sectors and gravity. Every structural claim is verified numerically to machine precision (residuals 10^{-14} – 10^{-16}) or as exact rationals, in a public test suite. An eleven-paper series gives the full construction.

Core claims (each a deviation from, or derivation of, the Standard Model)

1. **Light is a rotation rate, not a phase velocity** ($c_{\text{lat}} = d\Omega/d|\mathbf{k}| = 1/\sqrt{3}$). Maxwell’s curl law is its $O(\Omega)$ linearisation; energy conservation is geometric (length-preserving rotation).
2. **The photon is a bound pair of two spin- $1/2$ Weyl quanta** (de Broglie’s neutrino theory of light), not a fundamental spin-1 boson — massless, luminal, transverse, and **exactly non-birefringent**.
3. **Mass without a Higgs field**: a chiral- $SU(2)$ complex-mass step carries weak isospin as an exact gauge symmetry (Ward identity to 1.1×10^{-17}); the would-be Higgs direction is pure gauge. Hypercharge rides the same field.
4. **Exactly three fermion generations** — a theorem about the cubic point group O_h (maximal single-valued irrep T_{1u} is 3-dimensional; no 4-dimensional one exists), realised by the scalar mass selecting the odd-parity triplet. **A fourth is forbidden**.
5. **The Koide relation** $Q = \frac{2}{3}$ is an exact 45° equipartition of the cubic amplitude $\sqrt{m}$ ($Q(\phi) = 1/3 \cos^2 \phi$); electromagnetism selects the colour-free charged leptons to sit there.
6. **Confinement** is exact in 2D (area law, $\sigma = -\ln w(\beta) > 0$ for all β); in 3+1D it is a colour-dielectric dual superconductor cross-checked against gauge Monte-Carlo, governed by $\mathbb{Z}_3$ centre-phase closure.

7. **The Weinberg angle is derived:** $\sin^2 \theta_W = \frac{1}{4}$ from BCC swap geometry ($m_Z/m_W = 2/\sqrt{3}$, zero parameters).
8. **Gravity is one impedance-matched lattice dielectric** $K = e^{2GM/rc^2}$ (reciprocal lock $AB \equiv 1$): GR-identical PPN ($\beta = \gamma = 1$), and **Newton's constant is structural**, $G = a^2 c^3 / (8\pi \sqrt{3} \hbar)$, fixing $a/\ell_P = \sqrt{8\pi} 3^{1/4} = 6.5978$.
9. **Black holes are horizon-free** dielectric condensates (exponential metric): no event horizon, no standard Hawking glow, an information-paradox-free unitary substrate.

Headline verified numbers

Quantity	Model	Reference / status
c_{lat}	$1/\sqrt{3} = 0.5773503$	rotation-rate slope (axis & body diagonal)
$\sin^2 \theta_W \rightarrow m_Z/m_W$	$\frac{1}{4} \rightarrow 2/\sqrt{3} = 1.1547$	zero parameters; 1.77% from PDG
Koide Q	0.6666605	0.91σ from $\frac{2}{3}$; predicts $m_\tau = 1776.97 \text{ MeV}$ (6×10^{-5})
$a/\ell_P; G$	6.5978; 6.6743×10^{-11}	matches CODATA to 3×10^{-8}
BH shadow / Schwarzschild	$2e/3\sqrt{3} = 1.0463$	+4.63% (throat e , photon sphere $2\sqrt{e}$)
Deuteron E_b, κ	2.224 MeV, 0.2316 fm^{-1}	bound only by pion tensor force

Falsifiable predictions (with thresholds)

- **Photon-ring imaging.** The black-hole shadow is 4.63% **larger** than Schwarzschild (M87*: $41.5 \mu\text{as}$; Sgr A*: $55.7 \mu\text{as}$). A horizon-scale shadow excluding +4.63% at few-percent photon-ring precision (ngEHT / space-VLBI) **falsifies the model**.
- **Quantum-gravity dispersion.** A quadratic ($n = 2$) vacuum dispersion with energy scale $E_{\text{QG},2} = \sqrt{54} \hbar c / a \approx 1.36 \times 10^{19} \text{ GeV}$. **Any measured $n = 2$ time-of-flight bound above $1.36 \times 10^{19} \text{ GeV}$ kills the adopted lattice cell.**
- **Vacuum birefringence.** The physical (paired) photon is **exactly non-birefringent**; a confirmed first-order vacuum birefringence in GRB/AGN polarimetry would contradict it (and would, conversely, revive the excluded chiral construction).
- **Gravitational-wave echoes.** Horizon-free $\Rightarrow$ ringdowns reflect into **late-time echoes** (LIGO/Virgo/KAGRA, LISA). A confirmed true absorbing horizon with no echoes falsifies the picture.
- **Strong-field light bending.** Second-order deflection coefficient 4π (vs Schwarzschild $15\pi/4$); a measurement consistent with $15\pi/4$ but excluding 4π

falsifies it.

- **No event horizon / no thermal Hawking spectrum.** Detection of a thermal Hawking spectrum from an astrophysical horizon would contradict the model.
- **Mercury precession.** The nonlinear completion is fixed by $\beta = 1$ (42.98"/cy); the naive linear dielectric ($\beta = \frac{1}{2}$, 50.1"/cy) is already excluded — the model is committed to the exponential form.

Reproducibility

Every claim above runs in a public test suite to the stated residual; the lattice cell, couplings, and metric carry **no free fit parameters** in the structural sector. Source, tests, and per-finding write-ups accompany the papers.

*Full series (11 papers) and this summary: see the accompanying papers/ directory.
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